GPT Caddy

A utility to make GPT Golf easier.
See inside the run.
Decide what to change next.

I built it for myself — every GPT Golf run goes through it.

FORMAT

SLIDES · LIVE RUN · Q&A

EVERY RUN, END TO END

Launch → Monitor → **Diagnose** → Compare → Decide

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NEED

Naive GPT Golf is guess-and-check.

Tweak → wait → inspect loss → guess again?

Logs tell you what happened.

```
step:1000/20000 train_loss:3.0888 train_time:199487ms step_avg:199.48ms
step:1250/20000 train_loss:3.0824 train_time:249358ms step_avg:199.49ms
step:1500/20000 train_loss:3.0743 train_time:299232ms step_avg:199.49ms
step:1750/20000 train_loss:3.0707 train_time:349106ms step_avg:199.50ms
step:2000/20000 train_loss:3.0401 train_time:398978ms step_avg:199.49ms
step:2250/20000 train_loss:3.0157 train_time:448839ms step_avg:199.45ms
step:2500/20000 train_loss:2.9648 train_time:498704ms step_avg:199.46ms
step:2750/20000 train_loss:2.8870 train_time:548568ms step_avg:199.46ms
```

— not why.

SYSTEM

Shrink the loop. Iterate faster.

01 **Launch** rent + start → 02 **Monitor** live state → 03 **Diagnose** model health → 04 **Compare** run vs run → 05 **Decide** next change

Every pass through the loop should reduce uncertainty.

SYSTEM

Ephemeral GPU. Durable control plane.

RUNPOD GPU POD

Ephemeral compute

clone code · pull data
train · snapshot · terminate

Cloud. Stateless.

launch / control ←

→ events · logs · snapshots

GPT CADDY · LAPTOP

Durable control

launch · monitor · diagnose
store results · compare runs

Local. Persistent.

The pod can disappear. The durable run history survives.

SYSTEM

Time to first token matters.

SENTENCEPIECE FORK

6×+ faster

tokenization

>50GB Raw Text -> ~10B tokens -> 200MB Shards

Over 1 hour → 8-10 minutes

~50% less memory **~600% throughput**

Parallel buffering + divide-and-conquer the corpus.

CUDA / LOADER

<2GB Container

Faster starts.

Cut the stuff that isn’t used

20% the size of “full” distro

16 KB stubs satisfy the loader

Derived from the installed Torch binary.
Restore missing deps for special cases.

Multi-tier build. Bake CUDA/Torch once. Add new deps in seconds.

SYSTEM

A small event path keeps the live view honest.

POD
Event daemon
events · logs
snapshots

→

WIRE
Authenticated JSONL
HTTPS proxy
per-run token

→

LAPTOP
Event pipeline
persist · fan-out
diagnose

→

BROWSER
Live view
charts · cards
status

- **Boots before dependencies**
  stdlib-only daemon
- **Every event is durable**
  persisted to SQLite
- **One stream, three uses**
  browser · history · diagnostics

Pod → authenticate → laptop → browser / persistence / diagnostics / monitoring

LIVE

Watch three things move.

01

Race vs baseline

Ahead or behind at the same budget?

02

Cost in real time

Meter ticks; the pod stops itself.

03

Model health

Metrics live; probe runs off-GPU.

Keeps you informed without costing your focus.

DIAGNOSE

The diagnostics check their own arithmetic.

01 RE-COMPUTE layer outputs

02 RE-DERIVE loss independently

03 CROSS-CHECK BPB vs on-pod eval

gg-main-0622-205629 · main step 1000 just now WARN Deep Dive

WARN

3 checks need review in capacity and flow.

4 healthy 3 warnings 0 failing 0 unavailable 2 info

NEEDS ATTENTION

Dead neurons 7.2%
Units that never activate waste capacity; a large fraction means dead width. · warn ≥ 0.05, bad ≥ 0.2

Head redundancy 0.636
Attention heads computing near-identical outputs are redundant capacity. · warn ≥ 0.6, bad ≥ 0.85

Residual growth 0.375
Residual stream norm should grow moderately; too flat or exploding is unhealthy. · ok 0.5–4

Capacity 1 good · 2 warn
Dead neurons 7.2% Head redundancy 0.636
Attn entropy (min) 1.528

Flow 1 good · 1 warn · 1 info
Mean loss 2.778 Residual growth 0.375
Logit saturation 0.0%

Gradients 1 good
Grad norm ratio 11.436

Architecture 1 good · 1 info
Embed eff_rank 212.519 Zero-init escape (min) 4.281

1000 Latest

GPU trains. Laptop audits. Monitor progress without slowing down.

COMPARE

Did my change help?

01

Did loss improve?

curves + no-regression banner

02

Did validation improve?

BPB at equal progress

03

Did model health improve?

verdict + headline metrics

Compare at equal step or equal progress. Then decide what to change next.

Launch → Monitor → Diagnose → **Compare** → Decide ↺

CLOSE

From guess-and-check to informed decisions.

GPT Golf is an optimization problem without a ***profiler***.
Then I **built** one.

need → tool → evidence → decision

Compute is temporary but the memories last forever.